

Math 4441: Probability and Statistics

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Chapter 7
Parameter Estimation
(Part 1 – Point Estimation: Maximum Likelihood Estimation)
Lectures 25

Math 4441: Probability and Statistics

References:

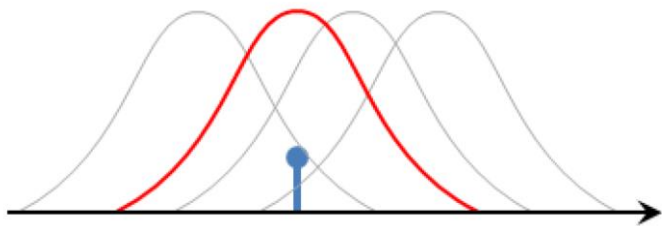
1. Goodman and Yates – Introduction to Probability and Stochastic Processes, 3rd Edition
2. Scott Miller and Donald Childers – Probability and Random Processes, 2nd Edition
3. Bertsekas and Tsitsiklis – Introduction to Probability, 2nd Edition

Parameter Estimation

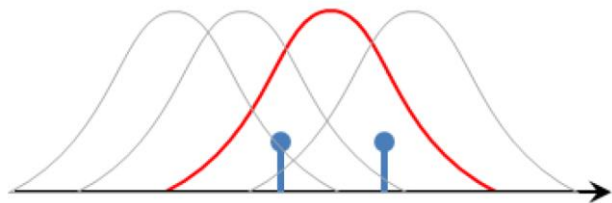
- $X_1, X_2, \dots, X_n$ random sample from distribution F_θ
 - θ is a set of unknown parameters
 - e.g., when F_θ is Poisson distributed with mean λ
 - $\theta = \{\lambda\}$
 - When F_θ is normal distributed with mean μ and variance σ^2
 - $\theta = \{\mu, \sigma^2\}$
 - In statistical inference,
 - A set of value is available as observed data
 - Inference is made about the unknown parameters

Maximum Likelihood Estimators

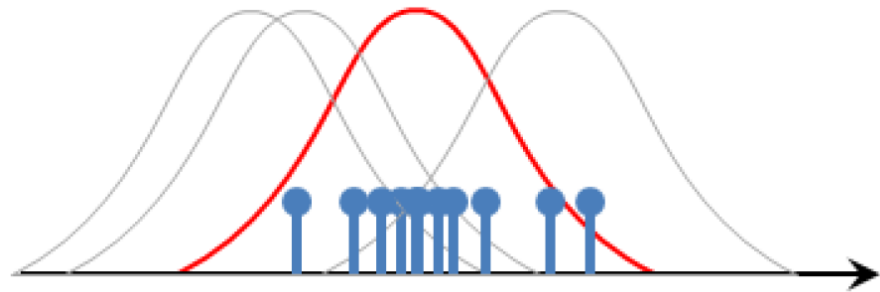
- Estimator: any statistic used to estimate the value of an (or a set of) unknown parameter θ
- Estimate: Observed value of the estimator is called the estimate
- MLE: Estimator that maximizes the probability of getting the observed data



- Which μ will give you the best Gaussian?
- When $\mu = x_1$, the probability of obtaining x_1 is the highest.



- Which μ will give you the best Gaussian?
- When $\mu = (x_1 + x_2)/2$, the prob. of obtaining x_1 and x_2 is highest.



- Which μ will give you the best Gaussian?
- When $\mu = \frac{1}{N} \sum_{n=1}^N x_n$, the prob. of obtaining $\{x_n\}$ is highest.

MLE of a Bernoulli Parameter

- n independent trials, each with success probability p
- What is the maximum likelihood estimator of p ?
- Observed data: $X_1, X_2, \dots, X_n$ where

$$X_i = \begin{cases} 1, & \text{if trial } i \text{ is a success} \\ 0, & \text{otherwise} \end{cases}$$

- $P[X_i = 1] = p = 1 - P[X_i = 0]$
- $P[X_i = x] = p^x(1-p)^{1-x}, \quad x = 0, 1$

MLE of a Bernoulli Parameter (Contd.)

- The likelihood of observing the data, i.e., the joint probability mass function of the data is

$$\begin{aligned}P[x_1, x_2, \dots, x_n] &= P[X_1 = x_1, X_2 = x_2, \dots, X_n = x_n] \\&= P[X_1 = x_1] \dots P[X_n = x_n] \\&= p^{x_1}(1-p)^{1-x_1} \dots p^{x_n}(1-p)^{1-x_n} \\&= p^{\sum_{i=1}^n x_i} (1-p)^{n-\sum_{i=1}^n x_i}\end{aligned}$$

- The log-likelihood function is

$$\ln P[x_1, \dots, x_n] = \sum_{i=1}^n x_i \ln p + (n - \sum_{i=1}^n x_i) \ln(1-p)$$

MLE of a Bernoulli Parameter (Contd.)

- Differentiating yields

$$\frac{d}{dp} \ln P[x_1, \dots, x_n] = \frac{\sum_{i=1}^n x_i}{p} - \frac{n - \sum_{i=1}^n x_i}{1 - p}$$

- Equating to zero and solving for p , gives $\hat{p}$

$$\frac{\sum_{i=1}^n x_i}{\hat{p}} = \frac{n - \sum_{i=1}^n x_i}{1 - \hat{p}}$$

$$\hat{p} = \frac{\sum_i x_i}{n}$$

MLE of the Poisson Parameter

- $X_1, X_2, \dots, X_n$ are independent Poisson random variables each with mean λ
- Determine the maximum likelihood estimation of λ
- The likelihood function is given by

$$L(\lambda) = P_{X_1, \dots, X_n}(x_1, \dots, x_n) = P_{X_1}(x_1) \dots P_{X_n}(x_n)$$

$$= \frac{e^{-\lambda} \lambda^{x_1}}{x_1!} \dots \frac{e^{-\lambda} \lambda^{x_n}}{x_n!} = \frac{e^{-n\lambda} \lambda^{\sum_{i=1}^n x_i}}{x_1! x_2! \dots x_n!}$$

$$LL(\lambda) = \ln L(\lambda) = -n\lambda + \sum_{i=1}^n x_i \ln \lambda - \ln \prod_{i=1}^n x_i!$$

MLE of the Poisson Parameter (Contd.)

- Differentiating both sides,

$$\frac{d}{d\lambda} LL(\lambda) = -n + \frac{\sum_{i=1}^n x_i}{\lambda}$$

- Equating to zero and solving for λ , gives $\hat{\lambda}$

$$\hat{\lambda} = \frac{\sum_{i=1}^n x_i}{n}$$

Example 7.2.d. The number of traffic accidents in Berkeley, California, in 10 randomly chosen nonrainy days in 1998 is as follows:

4, 0, 6, 5, 2, 1, 2, 0, 4, 3

Use these data to estimate the proportion of nonrainy days that had 2 or fewer accidents that year.

$$\overline{X} = \frac{1}{10} \sum_{i=1}^{10} X_i = 2.7$$

Since the long-run proportion of nonrainy days that have 2 or fewer accident is equal to $P\{X \leq 2\}$, where X is the random number of accidents in a day, i

$$e^{-2.7}(1 + 2.7 + (2.7)^2/2) = .4936$$

MLE in a Normal Population

- $X_1, X_2, \dots, X_n$ are IID $\sim N(\mu, \sigma^2)$
- $L(\mu, \sigma^2) = f_{X_1, \dots, X_n}(x_1, \dots, x_n)$

$$\begin{aligned} &= \prod_{i=1}^n \frac{1}{\sqrt{(2\pi)\sigma}} \exp\left[-\frac{(x_i - \mu)^2}{2\sigma^2}\right] \\ &= \left(\frac{1}{2\pi}\right)^{n/2} \frac{1}{\sigma^n} \exp\left[-\frac{\sum_{i=1}^n (x_i - \mu)^2}{2\sigma^2}\right] \end{aligned}$$

- The log-likelihood function is

$$LL(\mu, \sigma^2) = \ln L(\mu, \sigma^2) = -\frac{n}{2} \ln(2\pi) - n \ln \sigma - \frac{\sum_{i=1}^n (x_i - \mu)^2}{2\sigma^2}$$

MLE in a Normal Population (Contd.)

- Differentiating both sides wrt μ , and σ^2

$$\frac{\partial}{\partial \mu} l(\mu, \sigma^2) = \frac{\sum_{i=1}^n (x_i - \mu)}{\sigma^2}$$

$$\frac{\sum_{i=1}^n (x_i - \mu)}{\sigma^2} = 0 \gg \sum_i^n x_i - n\hat{\mu} = 0 \gg \hat{\mu} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\frac{\partial}{\partial \sigma} l(\mu, \sigma^2) = -\frac{n}{\sigma} + \frac{\sum_{i=1}^n (x_i - \mu)^2}{\sigma^3}$$

$$\hat{\sigma} = \left[\frac{\sum_i^n (x_i - \mu)}{n} \right]^{\frac{1}{2}}$$

Example 7.2.f. *Kolmogorov's law of fragmentation* states that the size of an individual particle in a large collection of particles resulting from the fragmentation of a mineral compound will have an approximate lognormal distribution,

Suppose that a sample of 10 grains of metallic sand taken from a large sand pile have respective lengths (in millimeters):

2.2, 3.4, 1.6, 0.8, 2.7, 3.3, 1.6, 2.8, 2.5, 1.9

Estimate the percentage of sand grains in the entire pile whose length is between 2 and 3 mm.

where a random variable X is said to have a *lognormal* distribution if $\log(X)$ has a normal distribution.

Solution. Taking the natural logarithm of these 10 data values, the following transformed data set results

.7885, 1.2238, .4700, $-.2231$, .9933, 1.1939, .4700, 1.0296, .9163, .6419

Because the sample mean and sample standard deviation of these data are

$$\bar{x} = .7504, \quad s = .4351$$

it follows that the logarithm of the length of a randomly chosen grain has a normal distribution with mean approximately equal to .7504 and with standard deviation approximately equal to .4351. Hence, if X is the length of the grain, then

$$\begin{aligned} P\{2 < X < 3\} &= P\{\log(2) < \log(X) < \log(3)\} \\ &= P\left\{\frac{\log(2) - .7504}{.4351} < \frac{\log(X) - .7504}{.4351} < \frac{\log(3) - .7504}{.4351}\right\} \end{aligned}$$

$$= P\left\{-.1316 < \frac{\log(X) - .7504}{.4351} < .8003\right\}$$

$$\approx \Phi(.8003) - \Phi(-.1316)$$

$$= .3405 \quad \blacksquare$$

Estimating Mean of Uniform Distribution

Example 7.2.g (Estimating the Mean of a Uniform Distribution). Suppose $X_1, \dots, X_n$ constitute a sample from a uniform distribution on $(0, \theta)$, where θ is unknown. Their joint density is thus

$$f(x_1, x_2, \dots, x_n | \theta) = \begin{cases} \frac{1}{\theta^n} & 0 < x_i < \theta, \quad i = 1, \dots, n \\ 0 & \text{otherwise} \end{cases}$$

This density is maximized by choosing θ as small as possible. Since θ must be at least as large as all of the observed values x_i , it follows that the smallest possible choice of θ is equal to $\max(x_1, x_2, \dots, x_n)$. Hence, the maximum likelihood estimator of θ is

$$\hat{\theta} = \max(X_1, X_2, \dots, X_n)$$

It easily follows from the foregoing that the maximum likelihood estimator of $\theta/2$, the mean of the distribution, is $\max(X_1, X_2, \dots, X_n)/2$. ■

